

Linear Algebra (English)

2024 Fall

Final Exam

2025.1.7

Exam Duration: 3 Hours

Name: _____

Student ID: _____

This exam includes 11 pages (including this page) and 5 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

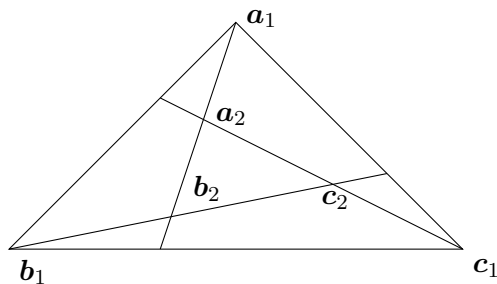
This exam is open book. You are allowed to consult your textbook and notes, and elementary calculators are fine (those without internet connections). Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. Any problem answered with nothing but a single answer would receive no credit. If you copied the result from pre-calculations (since we have an open portion for this exam), just write “by pre-calculation”. If you spot some results without any calculation, just write “I can spot that blah blah”.
- **Write cleanly and legible.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 50. If your total score exceeds 50 (there are 54 points in total), it will be recorded as 50.

Problem	Points	Score
1	10	
2	9	
3	12	
4	11	
5	12	
Total:	54	

1. Consider a triangle with vertices $\mathbf{a}_1, \mathbf{b}_1, \mathbf{c}_1 \in \mathbb{R}^2$. I can find trisection points on the three edges of the triangle, i.e., $\mathbf{x}_1 = \frac{2}{3}\mathbf{b}_1 + \frac{1}{3}\mathbf{c}_1, \mathbf{y}_1 = \frac{2}{3}\mathbf{c}_1 + \frac{1}{3}\mathbf{a}_1, \mathbf{z}_1 = \frac{2}{3}\mathbf{a}_1 + \frac{1}{3}\mathbf{b}_1$. Connect the line through $\mathbf{a}_1$ and $\mathbf{x}_1$, the line through $\mathbf{b}_1$ and $\mathbf{y}_1$, and the line through $\mathbf{c}_1$ and $\mathbf{z}_1$, and these three lines would intersect at three vertices $\mathbf{a}_2, \mathbf{b}_2, \mathbf{c}_2$ and form a smaller triangle. For a better idea, see the picture below.



In general, we want to study this process of going from $\mathbf{a}_n, \mathbf{b}_n, \mathbf{c}_n$ to $\mathbf{a}_{n+1}, \mathbf{b}_{n+1}, \mathbf{c}_{n+1}$. By high school geometry, it is possible to see that, for example,

$$\begin{aligned}\mathbf{a}_{n+1} &= \frac{4}{7}\mathbf{a}_n + \frac{2}{7}\mathbf{b}_n + \frac{1}{7}\mathbf{c}_n, \\ \mathbf{b}_{n+1} &= \frac{4}{7}\mathbf{b}_n + \frac{2}{7}\mathbf{c}_n + \frac{1}{7}\mathbf{a}_n, \\ \mathbf{c}_{n+1} &= \frac{4}{7}\mathbf{c}_n + \frac{2}{7}\mathbf{a}_n + \frac{1}{7}\mathbf{b}_n.\end{aligned}$$

Obviously the subproblems are going to analyze this dynamical system. Eigenvalues and diagonalizations are going to be important.

- (2 points) Find a matrix A such that $[\mathbf{a}_n \ \mathbf{b}_n \ \mathbf{c}_n] A = [\mathbf{a}_{n+1} \ \mathbf{b}_{n+1} \ \mathbf{c}_{n+1}]$, and check if A is a normal matrix or not.
- (2 points) The iteration process must be related to some rotation. Let $R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. This is a rotation on $\mathbb{R}^3$ around which line by how much? (Don't worry about clockwise or counter clockwise.)
- (2 points) Write A as a polynomial of R . Find all eigenvalues and eigenvectors of R and for A . (You may use w to represent $e^{\frac{2\pi}{3}i}$.)
- (2 points) Find $\lim \mathbf{a}_n, \lim \mathbf{b}_n, \lim \mathbf{c}_n$, and express your result as a linear combination of $\mathbf{a}_1, \mathbf{b}_1, \mathbf{c}_1$.
- (2 points) Suppose rather than choosing the trisection points, I pick the two fifth points. I.e., we choose $\mathbf{x}_1 = \frac{3}{5}\mathbf{b}_1 + \frac{2}{5}\mathbf{c}_1$, and so on for $\mathbf{y}_1, \mathbf{z}_1$, and use these to construct the smaller triangles $\mathbf{a}_2, \mathbf{b}_2, \mathbf{c}_2$. Then the new iteration formula is $\mathbf{a}_{n+1} = \frac{9}{19}\mathbf{a}_n + \frac{6}{19}\mathbf{b}_n + \frac{4}{19}\mathbf{c}_n$ and so on. Find $\lim \mathbf{a}_n, \lim \mathbf{b}_n, \lim \mathbf{c}_n$, and express your result as a linear combination of $\mathbf{a}_1, \mathbf{b}_1, \mathbf{c}_1$.

Answer:

1. $A = \frac{1}{7} \begin{bmatrix} 4 & 1 & 2 \\ 2 & 4 & 1 \\ 1 & 2 & 4 \end{bmatrix}$. A is normal.
2. This is a rotation around the line $x = y = z$ in $\mathbb{R}^3$ by $\frac{2\pi}{3}$.
3. $A = \frac{4}{7}I + \frac{1}{7}R + \frac{2}{7}R^2$. For R , the eigenvalues are $1, w, w^2$, and the eigenspaces are the span of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ w \\ w^2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ w^2 \\ w \end{bmatrix}$ respectively. For A , the eigenvalues are $1, \frac{4+w+2w^2}{7} = \frac{3+w^2}{7}, \frac{4+w^2+2w}{7} = \frac{3+w}{7}$, and the eigenspaces are the same as above.
4. All three limits are $\frac{a_1+b_1+c_1}{3}$.
5. All three limits are $\frac{a_1+b_1+c_1}{3}$. A is always going to be a Markov cyclic matrix with eigenvalue 1 and eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and this is the only thing that matters.

2. Suppose you have three classes, linear algebra, calculus, probability. Each class may provide you with conscious stress and unconscious stress.

Let's say if you put efforts a, b, c into linear algebra, calculus, probability respectively, then you get conscious stress x and unconscious stress y . Suppose this

process is linear, so we have $A \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ for a matrix A given as

$$A = \begin{bmatrix} 3 & -3 & -4 \\ 4 & 6 & 3 \end{bmatrix}.$$

Note that we define zero as “average”, so your effort values and stress values could be both positive or negative. The subproblems will analyze effort strategies and psychological impacts. Do SVD before hand!

- (a) (2 points) Suppose I want to minimize total stress with given amount of effort. Find a UNIT vector $\mathbf{x}$ such that $\|A\mathbf{x}\|$ is as small as possible, and also find this minimal value of $\|A\mathbf{x}\|$ under this condition. Which class would you most likely fail to reach this minimum?
- (b) (2 points) In general A would squash a sphere (in $\mathbb{R}^3$) into an ellipse (in $\mathbb{R}^2$). If we apply the map A to the unit sphere, what are the directions and lengths of the long half-axis and the short half-axis?
- (c) (3 points) Stress may cause your head to expand. (压力让人头大) Say I decide to drop a class, then let A_i be A deleting its i -th column. If S is the unit circle (my head), which one of $A_1(S)$, $A_2(S)$, $A_3(S)$ has the smallest area (your resulting head size)? For the sake of a smaller head, which class should you drop?
- (d) (2 points) Write A as the sum of two rank one matrices B, C , such that all columns of B are perpendicular to all columns of C , and all rows of B are perpendicular to all rows of C .

Answer: We have $A = U\Sigma V^T$. Here $U = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$, $\Sigma = \begin{bmatrix} \sqrt{70} & 0 & 0 \\ 0 & 5 & 0 \end{bmatrix}$, and

$$V = \begin{bmatrix} \frac{1}{\sqrt{14}} & \frac{2}{\sqrt{5}} & \frac{3}{\sqrt{70}} \\ \frac{3}{\sqrt{14}} & 0 & \frac{-5}{\sqrt{70}} \\ \frac{2}{\sqrt{14}} & \frac{-1}{\sqrt{5}} & \frac{6}{\sqrt{70}} \end{bmatrix}.$$

1. $\mathbf{x} = \frac{1}{\sqrt{70}} \begin{bmatrix} 3 \\ -5 \\ 6 \end{bmatrix}$, and the resulting $\|A\mathbf{x}\| = 0$. Most likely you will fail calculus.

Of course if you picked the opposite unit vector, then you will most likely fail probability.

2. The long half-axis has direction $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and length $\sqrt{70}$. The short half-axis has direction $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and length 5.

3. $\det(A_1) = 15, \det(A_2) = 25, \det(A_3) = 30$. You should drop linear algebra.

4. $B = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \end{bmatrix}$.

3. Let V be a function space with basis $\mathcal{B} = (e^{x+y}, x^2, xy, y^2)$, and let W be the space of 2×2 real matrices with basis $\mathcal{C} = (\mathbf{e}_1\mathbf{e}_1^T, \mathbf{e}_1\mathbf{e}_2^T, \mathbf{e}_2\mathbf{e}_1^T, \mathbf{e}_2\mathbf{e}_2^T)$. We have a map

$H : V \rightarrow W$ by sending each function $f \in V$ to its Hessian matrix $\begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$ at

the point $(x, y) = (0, 0)$. We also have a map $B : W \rightarrow V$ by sending each matrix

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ to the function $f_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = ax^2 + (b+c)xy + dy^2$.

The subproblems will analyze the behaviors of B, H and their potential compositions.

- (a) (3 points) Under the standard matrix inner product (i.e., $\langle A, B \rangle = \text{trace}(A^T B)$), is basis $\mathcal{C}$ orthonormal?

- (b) (2 points) Find the matrices for H and for B under the given basis.

- (c) (2 points) Show that $\frac{1}{2}HB$ is idempotent. Under the standard inner product W , is this an orthogonal projection?

- (d) (3 points) Find all eigenvectors and eigenvalues of HB . (You may use coordinates to calculate or not. But for the answer, please describe eigenvectors as matrices in W rather than coordinate vectors under $\mathcal{C}$.)

- (e) (2 points) Let $\mathcal{B}' = ((x+y)^2, (x-y)^2, x^2 - y^2, e^{x+y})$, find the change of coordinate matrix $I_{\mathcal{B} \leftarrow \mathcal{B}'}$ and the matrix $H_{\mathcal{C} \leftarrow \mathcal{B}'}$.

Answer:

1. $\mathcal{C}$ is ONB.

2. $H = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

3. $\frac{1}{2}HB = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$. Easy to verify that $(HB)^2 = 2HB$ and it is obvi-

ously symmetric. Since this is over ONB $\mathcal{C}$, therefore this is an orthogonal projection.

4. The eigenvalues are 2, 2, 2, 0, and the eigenspaces are the symmetric matrices and the skew symmetric matrices.

5. We have $I_{\mathcal{B} \leftarrow \mathcal{B}'} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 2 & -2 & 0 & 0 \\ 1 & 1 & -1 & 0 \end{bmatrix}$ and $H_{\mathcal{C} \leftarrow \mathcal{B}'} = \begin{bmatrix} 2 & 2 & 2 & 1 \\ 2 & -2 & 0 & 1 \\ 2 & -2 & 0 & 1 \\ 2 & 2 & -2 & 1 \end{bmatrix}$.

4. What would happen if a matrix is both Hermitian and Unitary? Using this fact, you can actually deduce lots of eigen-information already. The subproblems will talk about a related matrix $H = \begin{bmatrix} 1 & i \\ -i & -1 \end{bmatrix}$ and various block matrices using H .

(a) (3 points) For different values of t , find the LDU decomposition of $H + \begin{bmatrix} 0 & 0 \\ 0 & t \end{bmatrix}$.

(b) (2 points) For different values of t , what is the definiteness of $H + \begin{bmatrix} 0 & 0 \\ 0 & t \end{bmatrix}$?

(c) (3 points) For different values of t , what is the definiteness of $H + tI$?

(d) (3 points) Let $M = \begin{bmatrix} H & 2H \\ 2H & 5H \end{bmatrix}$, find a lower triangular X such that $M = XDX^*$, where D is the congruence canonical form of M . (Hint: maybe write X in block forms and each block is lower triangular.)

Answer:

1. $\begin{bmatrix} 1 & i \\ -i & -1+t \end{bmatrix} = \begin{bmatrix} 1 & \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & -2+t \end{bmatrix} \begin{bmatrix} 1 & i \\ & 1 \end{bmatrix}.$
2. If $t > 2$ it is positive definite, if $t \geq 2$ it is positive semidefinite (say $t = 2$ is also OK). If $t < 2$ it is indefinite.
3. H has eigenvalues $\sqrt{2}, -\sqrt{2}$. So $H + tI$ is positive definite if $t > \sqrt{2}$, positive semidefinite if $t \geq \sqrt{2}$, indefinite if $\sqrt{2} > t > -\sqrt{2}$, negative semidefinite if $t \leq -\sqrt{2}$, and negative definite if $t < -\sqrt{2}$.

$$4. H = \begin{bmatrix} 1 & \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & -2 \end{bmatrix} \begin{bmatrix} 1 & i \\ & 1 \end{bmatrix}. \text{ Let } L = \begin{bmatrix} 1 & \\ i & \sqrt{2} \end{bmatrix}, \text{ then } H = L \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} L^*.$$

Now for the block matrix, $M = \begin{bmatrix} H & 2H \\ 2H & 5H \end{bmatrix} = \begin{bmatrix} I & \\ 2I & I \end{bmatrix} \begin{bmatrix} H & \\ & H \end{bmatrix} \begin{bmatrix} I & 2I \\ & I \end{bmatrix}$. So

$$X = \begin{bmatrix} L & \\ 2L & L \end{bmatrix}. \text{ We have } D = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{bmatrix}.$$

5. You are supposed to receive a linear combination of two signals, say the two signals corresponds to $\mathbf{a}_1$ and $\mathbf{a}_2$. The signal you actually receive is $\mathbf{b}$, and we aim to decompose $\mathbf{b}$ as linear combinations of $\mathbf{a}_1, \mathbf{a}_2$, and find the coefficients for $\mathbf{a}_1$ and $\mathbf{a}_2$.

However, if you try to do this decomposition, you might see something fishy. Sometimes $\mathbf{b}$ might involve some random noises that are not supposed to be part of the signal!

The subproblems will attempt to filter away the noise, and reconstruct the correct signal which is clear of noise. We are going for an application of least square solutions.

- (a) (3 points) Say $\mathbf{a}_1 = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$ for all the subproblems. Is there any noise in $\mathbf{b}$? Also please find the LDU decomposition for the augmented matrix.

- (b) (3 points) For the matrix $A = [\mathbf{a}_1 \ \mathbf{a}_2]$, do Gram-Schmidt decomposition $A = QR$, and find the Cholesky decomposition $A^T A = LL^T$.

- (c) (3 points) Find the unit normal vector perpendicular to $\text{Ran}(A)$, and find the projection matrix P to the range of A , and calculate $P\mathbf{b}$ when $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$, which is the most likely actual signal without noise.

- (d) (3 points) Recall the Cholesky decomposition $A^T A = LL^T$ in the second subproblem. For $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$, find $\mathbf{y}$ such that $L\mathbf{y} = A^T \mathbf{b}$, and then find $\mathbf{x}$ such that $L^T \mathbf{x} = \mathbf{y}$. Is the result $\mathbf{x}$ the least square solution?

Answer:

1. Yes, since there is no solution to $A\mathbf{x} = \mathbf{b}$. $\begin{bmatrix} 3 & 3 & 3 \\ 3 & -1 & 1 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 3 & & \\ & -4 & \\ & & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ & 1 & \frac{1}{2} \\ & & 1 \end{bmatrix}$.
2. $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ & 2 \end{bmatrix}$, so $Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 \end{bmatrix}$ and $R = \begin{bmatrix} 3\sqrt{3} & \sqrt{3} \\ & 2\sqrt{2} \end{bmatrix}$, and $L = R^T$.
3. The unit normal vector is $\frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$, and $P = \frac{1}{6} \begin{bmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$, and $P\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$.
4. $L\mathbf{y} = A^T\mathbf{b} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$, so $\mathbf{y} = \begin{bmatrix} \sqrt{3} \\ \sqrt{2} \end{bmatrix}$, and thus $\mathbf{x} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{2} \end{bmatrix}$. This is the least square solution.